

Multi-class classification

Approach to multi-class

A natural **extension of binary classification**

- The **aim** is to **assign a discrete label** to examples
- $\neq$ Binary classification: **you have $K > 2$ classes** to choose from

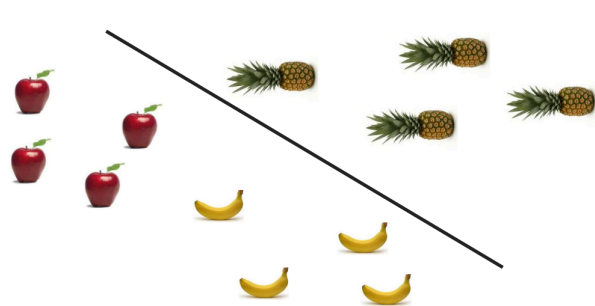
We can **modify the implementation of already seen algorithms**

- KNN: no algorithmic changes
- **Perceptron**: it is **hard** to **separate 3 or more classes** with just **one line** $\rightarrow$ need to change it

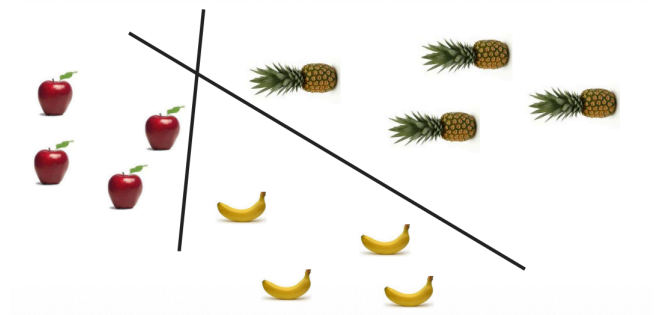
Binary to multi-class classification

Given a generic binary classifier we can **solve a multi-class problem**

- **Applying** the **binary classifier subsequent times**



Binary classification in perceptron



Multi-class classification in perceptron

2 approaches

- **OVA** (= One vs All)
- **AVA** (= All vs All)

OVA - One vs All

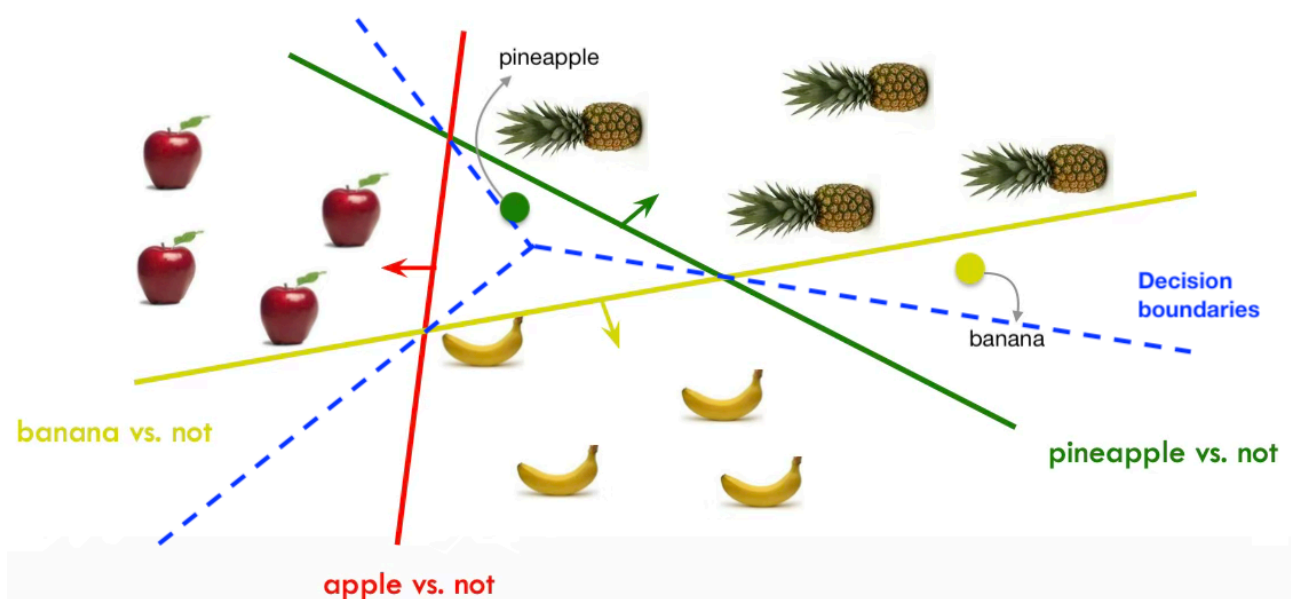
For each label $k = 1, \dots, K$ a binary problem is defined

- Examples with label $k \rightarrow$ are **positive**
- **Other** examples $\rightarrow$ are **negative**

Training set		Apple vs Not	Banana vs Not	Pineapple vs Not
 apple		1	-1	-1
 pineapple		-1	-1	1
 apple		1	-1	-1
 banana		-1	1	-1
 banana		-1	1	-1

How to classify?

- **Pick the most confident positive**
- If none, pick the least confident negative



OVA formal algorithms

Algorithm 1 OneVersusAllTrain($D^{multiclass}$, BinaryTrain)

```

1: for  $i = 1$  to  $K$  do
2:    $D^{bin} \leftarrow$  relabel  $D^{multiclass}$  so class  $i$  is positive and  $-i$  is negative
3:    $f_i \leftarrow$  BinaryTrain( $D^{bin}$ )
4: end for
5: return  $f_1, \dots, f_K$ 

```

Algorithm 2 OneVersusAllTest($f_1, \dots, f_K, \hat{x}$)

```

1:  $score \leftarrow \langle 0, 0, \dots, 0 \rangle$  ▷ initialize  $K$ -many scores to zero
2: for  $i = 1$  to  $K$  do
3:    $y \leftarrow f_i(\hat{x})$ 
4:    $score_i \leftarrow score_i + y$ 
5: end for
6: return  $\arg \max_k score_k$ 

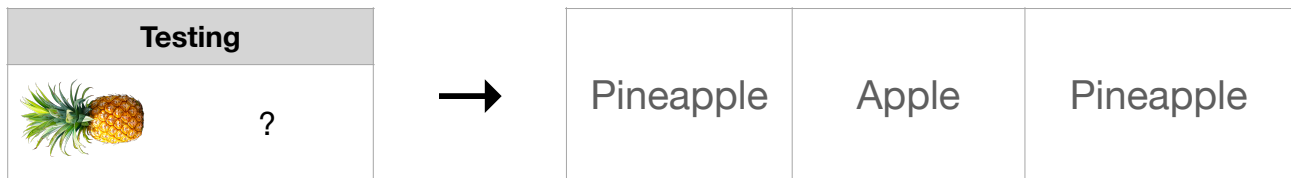
```

AVA - All vs All

For each possible couple combination (i, j) of the K classes a binary classifier is trained

- Only on the examples of those two classes
- In total $K(K - 1)/2$ classifiers are trained

Training set		Apple vs Pineapple	Apple vs Banana	Pineapple vs Banana
 apple		1	1	–
 pineapple		-1	–	1
 apple	→	1	1	–
 banana		–	-1	-1
 banana		–	-1	-1



How do we **classify**? A few options:

- Take a **majority vote**
- Take a **weighted vote based on confidence**
 - y is positive and was classified as of type j
 - Raise the score for j
 - Lower the score for i
 - y is negative and was classified as of type i
 - Lower the score for j
 - Raise the score for i

AVA training pseudocode

Algorithm 1 AVA Training

```

1: for  $i = 1$  to number of labels do
2:   for  $j = i + 1$  to number of labels do
3:     train a classifier  $F_{ij}$  to distinguish between  $label_j$  and  $label_i$ 
4:     create a dataset with all examples with  $label_j$  labeled positive
5:     and all examples with  $label_i$  labeled negative
6:     train classifier  $F_{ij}$  on this subset of the data
7:   end for
8: end for

```

AVA formal algorithms

Algorithm 1 AllVersusAllTrain($D^{multiclass}$, BinaryTrain)

```

1:  $f_{ij} \leftarrow \emptyset, \forall 1 \leq i < j \leq K$ 
2: for  $i = 1$  to  $K - 1$  do
3:    $D^{pos} \leftarrow$  all  $x \in D^{multiclass}$  labeled  $i$ 
4:   for  $j = i + 1$  to  $K$  do
5:      $D^{neg} \leftarrow$  all  $x \in D^{multiclass}$  labeled  $j$ 
6:      $D^{bin} \leftarrow \{(x, +1) : x \in D^{pos}\} \cup \{(x, -1) : x \in D^{neg}\}$ 
7:      $f_{ij} \leftarrow \text{BinaryTrain}(D^{bin})$ 
8:   end for
9: end for
10: return all  $f_{ij}$ s

```

Algorithm 2 AllVersusAllTest(all f_{ij} , $\hat{x}$)

```

1:  $score \leftarrow \langle 0, 0, \dots, 0 \rangle$  ▷ initialize  $K$ -many scores to zero
2: for  $i = 1$  to  $K - 1$  do
3:   for  $j = i + 1$  to  $K$  do
4:      $y \leftarrow f_{ij}(\hat{x})$ 
5:      $score_i \leftarrow score_i + y$ 
6:      $score_j \leftarrow score_j - y$ 
7:   end for
8: end for
9: return  $\arg \max_k score_k$ 

```

OVA vs AVA

Training time

- **AVA** learns more classifier each trained on much smaller data → **Faster** (if labels are equally balanced)

Testing time

- **AVA** has more classifiers → **Slower**

Error

- **AVA** trains on more balanced datasets
- **AVA** tests with more classifiers → more chances for error



The **most common approach** is **OVA**, or using a classifier that allows for multiple labels

Evaluation

Given some test examples and their prediction, **we want to measure the efficacy of a model**

Microaveraging and Macroaveraging

- **Microaveraging**: average over **examples**
- **Macroaveraging**: calculate the **evaluation score for each label**, then the **average** over labels

Testing set	Label	Prediction
	apple	pineapple
	pineapple	pineapple
	apple	apple
	banana	banana
	banana	pineapple

Microaveraging accuracy:

$$\frac{1 + 1 + 1}{5} = \frac{3}{5}$$

Macroaveraging accuracy:

$$\frac{\overbrace{1/2}^{\text{apple}} + \overbrace{1/1}^{\text{pineapple}} + \overbrace{1/2}^{\text{banana}}}{3} = \frac{2}{3}$$

Confusion matrix

Each entry (i, j) is the number of examples with label i predicted to have label j